

Week 5: Kinematics Problem-Solving Session

Instructor Notes and Worked Solutions

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1 Purpose

This Week 5 document replaces the usual theory notes with a guided problem-solving session covering the core manipulator-kinematics material from Weeks 3 and 4:

- forward kinematics with DH tables
- geometric forward kinematics for articulated arms
- ZYZ Euler angle extraction (previews the wrist sub-problem)

- analytical inverse kinematics for planar, SCARA, and elbow robots
- kinematic decoupling for a spherical wrist

The original numerical inverse-kinematics theory files were preserved in:

weeks/W05_numerical-inverse-kinematics/theory/archive/numerical-ik-original/

2 Session Structure

2.1 Recommended Timing

Problem	Topic	Time
1	2R planar FK with DH	15 min
2	SCARA FK	20 min
3	Spherical wrist FK + ZYZ extraction	25 min
4	3R elbow-arm FK position	20 min
5	2R planar IK	20 min
6	SCARA IK (inverse of Problem 2)	25 min
7	3R elbow-arm IK	25 min
8	6-DOF wrist decoupling	30 min

2.2 FK/IK Pairings

Three FK/IK problem pairs reinforce the forward-inverse connection:

- Problems 2 and 6: SCARA FK $\rightarrow$ SCARA IK (same robot, same target)
- Problems 4 and 7: 3R elbow FK $\rightarrow$ 3R elbow IK (same robot, same target)
- Problems 3 and 8: ZYZ forward $\rightarrow$ ZYZ extraction (same formulas)

Explicitly point out these connections when transitioning between Part A and Part B.

2.3 Recurring Checklist

For each problem, students should be pushed to follow the same routine:

1. Sketch the robot and name the joint variables.
2. Decide whether a full homogeneous transform is necessary.
3. Identify the minimum geometry needed.
4. Use `atan2` instead of `atan`.
5. Verify the final result with a forward check.

3 Problem 1: 2R Planar Forward Kinematics

3.1 Statement

A 2-link planar robot has link lengths $L_1 = 0.8$ m and $L_2 = 0.6$ m. At one instant,

$$\theta_1 = 30^\circ, \quad \theta_2 = 45^\circ$$

Find:

1. the DH table
2. the two link transforms
3. the total transform T_2^0

4. the end-effector position and orientation

y

Problem 1: 2R planar FK

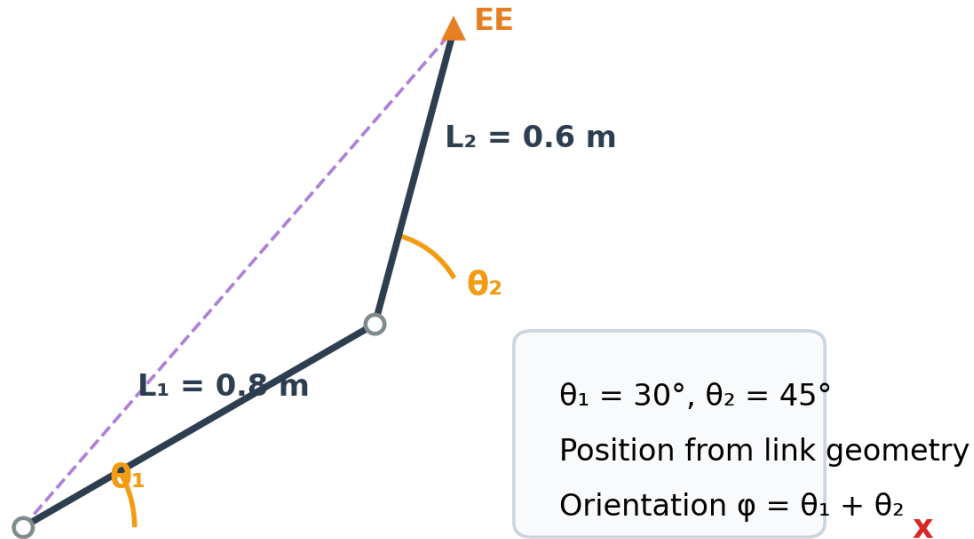


Figure 1: Problem 1 geometry

3.2 Solution

3.2.1 DH Table

Link	a_i	α_i	d_i	θ_i
1	0.8	0°	0	θ_1
2	0.6	0°	0	θ_2

Since both joints are revolute and the robot is planar:

- all motion is in the xy plane
- all z axes are parallel
- all $\alpha_i = 0$
- all $d_i = 0$

3.2.2 Link Transforms

$$T_1^0 = \begin{bmatrix} \cos 30^\circ & -\sin 30^\circ & 0 & 0.8 \cos 30^\circ \\ \sin 30^\circ & \cos 30^\circ & 0 & 0.8 \sin 30^\circ \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_2^1 = \begin{bmatrix} \cos 45^\circ & -\sin 45^\circ & 0 & 0.6 \cos 45^\circ \\ \sin 45^\circ & \cos 45^\circ & 0 & 0.6 \sin 45^\circ \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

3.2.3 Total Transform

Using the standard 2R planar result,

$$T_2^0 = \begin{bmatrix} \cos(\theta_1 + \theta_2) & -\sin(\theta_1 + \theta_2) & 0 & L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2) \\ \sin(\theta_1 + \theta_2) & \cos(\theta_1 + \theta_2) & 0 & L_1 \sin \theta_1 + L_2 \sin(\theta_1 + \theta_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Substituting the values:

$$T_2^0 = \begin{bmatrix} 0.2588 & -0.9659 & 0 & 0.8481 \\ 0.9659 & 0.2588 & 0 & 0.9796 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

3.2.4 Final Answer

$$\mathbf{p}_{ee} = \begin{bmatrix} 0.8481 \\ 0.9796 \\ 0 \end{bmatrix} \text{ m}, \quad \phi = 75^\circ$$

3.3 Teaching Note

This is the shortest problem of the set. It is useful for resetting notation before moving to SCARA and articulated-arm cases.

4 Problem 2: SCARA Forward Kinematics

4.1 Statement

Use the Week 3 SCARA convention. Given:

$$a_1 = 0.40 \text{ m}, \quad a_2 = 0.30 \text{ m}, \quad d_3 = 0.12 \text{ m}, \quad d_4 = 0.08 \text{ m}$$

$$\theta_1 = 20^\circ, \quad \theta_2 = 40^\circ, \quad \theta_4 = -30^\circ$$

Find:

1. the DH table
2. the end-effector position
3. the net tool orientation

4. the direction of the tool z -axis

Problem 2: SCARA geometry

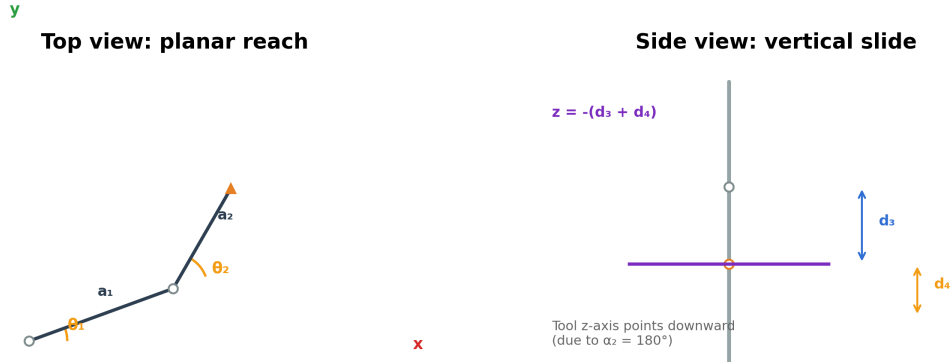


Figure 2: Problem 2 geometry

4.2 Solution

4.2.1 DH Table

Link	a_i	α_i	d_i	θ_i
1	0.40	0°	0	θ_1
2	0.30	180°	0	θ_2
3	0	0°	d_3	0
4	0	0°	d_4	θ_4

The key geometric feature is $\alpha_2 = 180^\circ$, which flips the z -axis between links 2 and 3.

4.2.2 Position

From the SCARA forward-kinematics result:

$$x = a_1 \cos \theta_1 + a_2 \cos(\theta_1 + \theta_2)$$

$$y = a_1 \sin \theta_1 + a_2 \sin(\theta_1 + \theta_2)$$

$$z = -(d_3 + d_4)$$

Substitute:

$$x = 0.40 \cos 20^\circ + 0.30 \cos 60^\circ = 0.5259 \text{ m}$$

$$y = 0.40 \sin 20^\circ + 0.30 \sin 60^\circ = 0.3966 \text{ m}$$

$$z = -(0.12 + 0.08) = -0.20 \text{ m}$$

So,

$$\mathbf{p}_{ee} = \begin{bmatrix} 0.5259 \\ 0.3966 \\ -0.2000 \end{bmatrix} \text{ m}$$

4.2.3 Orientation

The planar rotation seen in the Week 3 SCARA result is:

$$\phi = \theta_1 + \theta_2 + \theta_4 = 30^\circ$$

Because $\alpha_2 = 180^\circ$ flips the z -axis between links 2 and 3, the final tool z -axis points downward ($-z_0$ direction).

4.3 Teaching Note

Students often recover the correct x and y but forget the sign on the z direction. Push them to trace the α twist through the chain rather than just memorize the formula. Ask: “What does $\alpha_2 = 180^\circ$ physically do?”

5 Problem 3: Spherical Wrist Orientation and ZYZ Extraction

5.1 Statement

For a spherical wrist:

$$R_6^3 = R_z(\theta_4)R_y(\theta_5)R_z(\theta_6)$$

Given:

$$\theta_4 = 30^\circ, \quad \theta_5 = 45^\circ, \quad \theta_6 = -60^\circ$$

Find:

1. the rotation matrix R_6^3
2. the approach vector (third column)
3. the geometric meaning of a spherical wrist
4. recover $\theta_4, \theta_5, \theta_6$ from R_6^3 using the ZYZ extraction formulas

Problem 3: spherical wrist

$$R_6^3 = R_z(\theta_4) R_y(\theta_5) R_z(\theta_6)$$

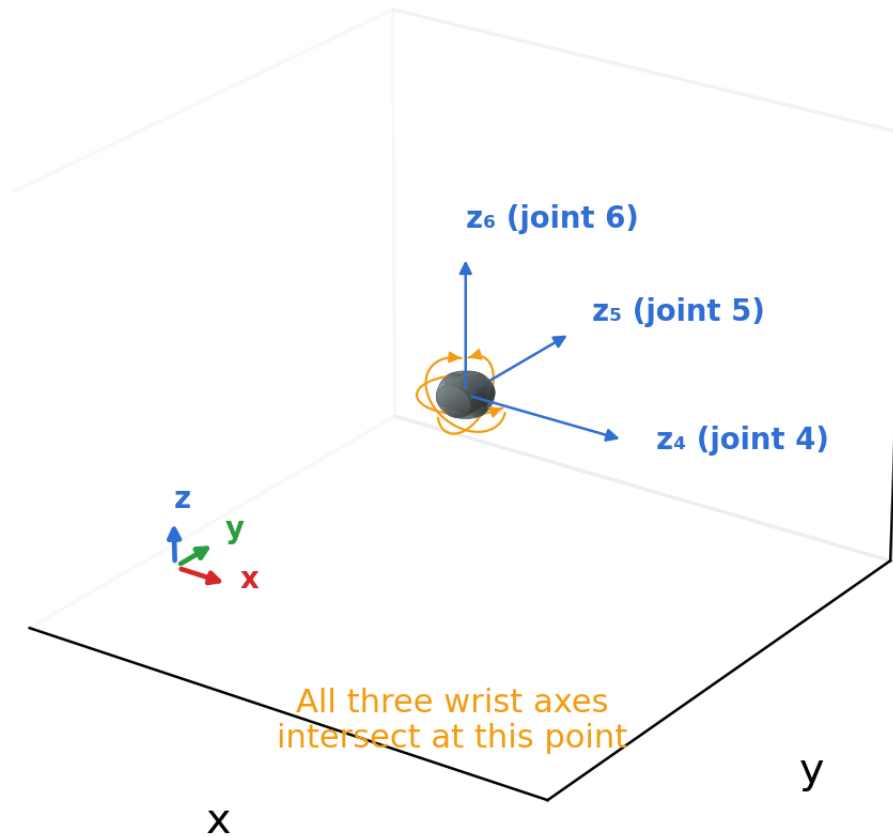


Figure 3: Problem 3 geometry

5.2 Solution

5.2.1 Matrix Multiplication

After multiplying the three rotations,

$$R_6^3 = \begin{bmatrix} 0.7392 & 0.2803 & 0.6124 \\ -0.5732 & 0.7392 & 0.3536 \\ -0.3536 & -0.6124 & 0.7071 \end{bmatrix}$$

5.2.2 Approach Vector

The third column is the tool approach vector:

$$\mathbf{a} = \begin{bmatrix} 0.6124 \\ 0.3536 \\ 0.7071 \end{bmatrix}$$

5.2.3 Interpretation

The wrist is called spherical because:

- joints 4, 5, and 6 intersect at one common point
- those joints can reorient the tool without changing the wrist-center location
- this makes inverse kinematics separable into position and orientation sub-problems

5.2.4 ZYZ Extraction

Let r_{ij} denote the (i, j) entry of R_6^3 . The standard ZYZ extraction formulas are:

$$\theta_5 = \text{atan2}\left(\sqrt{r_{13}^2 + r_{23}^2}, r_{33}\right) = \text{atan2}(0.7071, 0.7071) = 45^\circ$$

$$\theta_4 = \text{atan2}(r_{23}, r_{13}) = \text{atan2}(0.3536, 0.6124) = 30^\circ$$

$$\theta_6 = \text{atan2}(r_{32}, -r_{31}) = \text{atan2}(-0.6124, 0.3536) = -60^\circ$$

The recovered angles match the original input, confirming both the matrix computation and the extraction formulas.

5.3 Teaching Note

This problem now has two purposes: (1) build the wrist rotation matrix, and (2) practice the ZYZ extraction that students will reuse in Problem 8. Point out explicitly that the $\theta_5, \theta_4, \theta_6$ extraction is exactly the same procedure they'll apply to R_6^3 in the decoupling problem. If time is tight, let students do the forward multiplication as a group and focus individual work on the extraction.

6 Problem 4: 3R Elbow-Arm Forward Kinematics

6.1 Statement

For a 3R elbow robot:

$$d_1 = 0.30 \text{ m}, \quad a_2 = 0.40 \text{ m}, \quad a_3 = 0.30 \text{ m}$$

$$\theta_1 = 45^\circ, \quad \theta_2 = 30^\circ, \quad \theta_3 = -60^\circ$$

Find the end-effector position using the geometric decomposition:

$$r = a_2 \cos \theta_2 + a_3 \cos(\theta_2 + \theta_3)$$

Problem 4: 3R elbow-arm FK

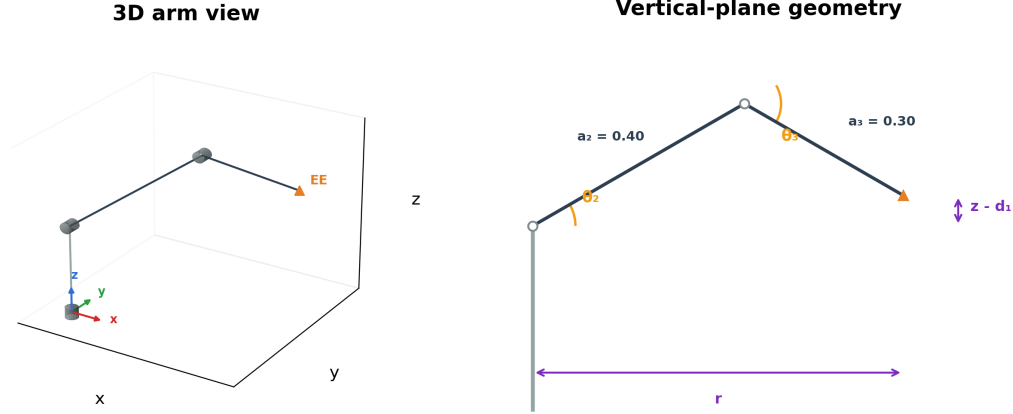


Figure 4: Problem 4 geometry

6.2 Solution

6.2.1 Radial Reach

$$r = 0.40 \cos 30^\circ + 0.30 \cos(-30^\circ) = 0.3464 + 0.2598 = 0.6062 \text{ m}$$

6.2.2 Cartesian Position

$$x = r \cos \theta_1 = 0.6062 \cos 45^\circ = 0.4287 \text{ m}$$

$$y = r \sin \theta_1 = 0.6062 \sin 45^\circ = 0.4287 \text{ m}$$

$$z = d_1 + a_2 \sin \theta_2 + a_3 \sin(\theta_2 + \theta_3)$$

$$z = 0.30 + 0.40 \sin 30^\circ + 0.30 \sin(-30^\circ) = 0.30 + 0.20 - 0.15 = 0.35 \text{ m}$$

Final answer:

$$\mathbf{p}_{ee} = \begin{bmatrix} 0.4287 \\ 0.4287 \\ 0.3500 \end{bmatrix} \text{ m}$$

6.2.3 Why the Decomposition Matters

θ_1 only rotates the vertical plane around z_0 . Once θ_1 is fixed, the remaining 2-link chain (a_2, a_3) lives in a single plane. This same geometry reappears in IK (Problem 7), where we first solve θ_1 , then reduce to the vertical-plane 2-link problem.

6.3 Teaching Note

This is the forward version of Problem 7. It is worth explicitly pointing out that the same triangle appears again in inverse form. Ask students: “If I give you the end-effector position, can you recover these angles?”

7 Problem 5: 2R Planar Inverse Kinematics

7.1 Statement

A 2-link planar arm has equal lengths:

$$L_1 = 0.50 \text{ m}, \quad L_2 = 0.50 \text{ m}$$

Target:

$$\mathbf{p}_{target} = \begin{bmatrix} 0.50 \\ 0.50 \end{bmatrix} \text{ m}$$

Find both IK solutions and label them using the Week 4 convention ($+\theta_2 = \text{elbow-up}$).

Problem 5: 2R planar IK

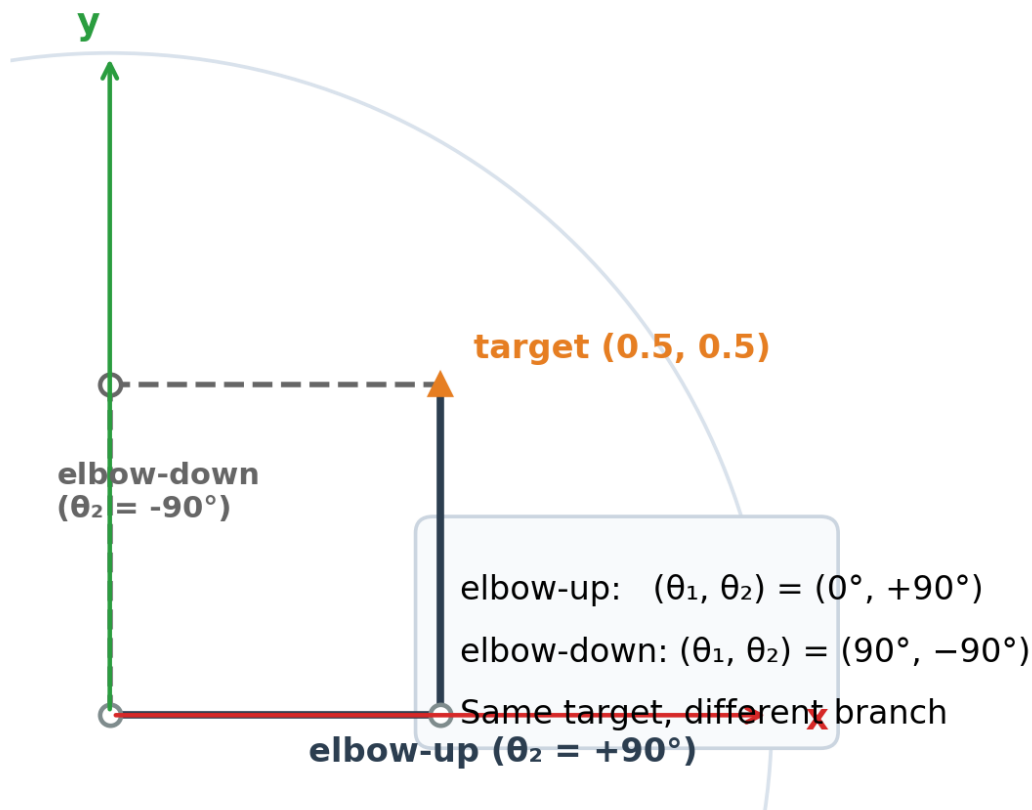


Figure 5: Problem 5 geometry

7.2 Solution

7.2.1 Reachability Check

The target radius is:

$$r = \sqrt{0.50^2 + 0.50^2} = 0.7071 \text{ m}$$

The arm workspace satisfies:

$$|L_1 - L_2| \leq r \leq L_1 + L_2$$

$$0 \leq 0.7071 \leq 1.0$$

So the target is reachable.

7.2.2 Solve for θ_2

$$\cos \theta_2 = \frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1 L_2}$$

Substitute:

$$\cos \theta_2 = \frac{0.25 + 0.25 - 0.25 - 0.25}{2(0.50)(0.50)} = 0$$

Hence:

$$\theta_2 = \pm 90^\circ$$

7.2.3 Solve for θ_1

Use:

$$\theta_1 = \text{atan2}(y, x) - \text{atan2}(L_2 \sin \theta_2, L_1 + L_2 \cos \theta_2)$$

Since $\text{atan2}(0.5, 0.5) = 45^\circ$:

For $\theta_2 = 90^\circ$ (elbow-up):

$$\theta_1 = 45^\circ - \text{atan2}(0.5, 0.5) = 0^\circ$$

For $\theta_2 = -90^\circ$ (elbow-down):

$$\theta_1 = 45^\circ - \text{atan2}(-0.5, 0.5) = 90^\circ$$

7.2.4 FK Verification

For elbow-up $(0^\circ, 90^\circ)$:

$$x = 0.5 \cos 0^\circ + 0.5 \cos 90^\circ = 0.5 \checkmark$$

$$y = 0.5 \sin 0^\circ + 0.5 \sin 90^\circ = 0.5 \checkmark$$

For elbow-down $(90^\circ, -90^\circ)$:

$$x = 0.5 \cos 90^\circ + 0.5 \cos 0^\circ = 0.5 \checkmark$$

$$y = 0.5 \sin 90^\circ + 0.5 \sin 0^\circ = 0.5 \checkmark$$

7.2.5 Final Answers

- **Elbow-up** ($\theta_2 > 0$): $(\theta_1, \theta_2) = (0^\circ, 90^\circ)$
- **Elbow-down** ($\theta_2 < 0$): $(\theta_1, \theta_2) = (90^\circ, -90^\circ)$

7.3 Teaching Note

Emphasize that the Week 4 convention defines elbow-up as $\theta_2 > 0$ (positive sine branch) and elbow-down as $\theta_2 < 0$ (negative sine branch). This is consistent with the `ik_2link()` function from Week 4 where `elbow='up'` selects $s_2 = +\sqrt{1-c_2^2}$.

8 Problem 6: SCARA Inverse Kinematics

8.1 Statement

Use the same SCARA robot as Problem 2:

$$a_1 = 0.40 \text{ m}, \quad a_2 = 0.30 \text{ m}, \quad d_4 = 0.08 \text{ m (fixed)}$$

Target pose:

$$\mathbf{p}_{target} = \begin{bmatrix} 0.5259 \\ 0.3966 \\ -0.2000 \end{bmatrix} \text{ m}, \quad \phi_{target} = 30^\circ$$

Find all joint variables for both elbow branches.

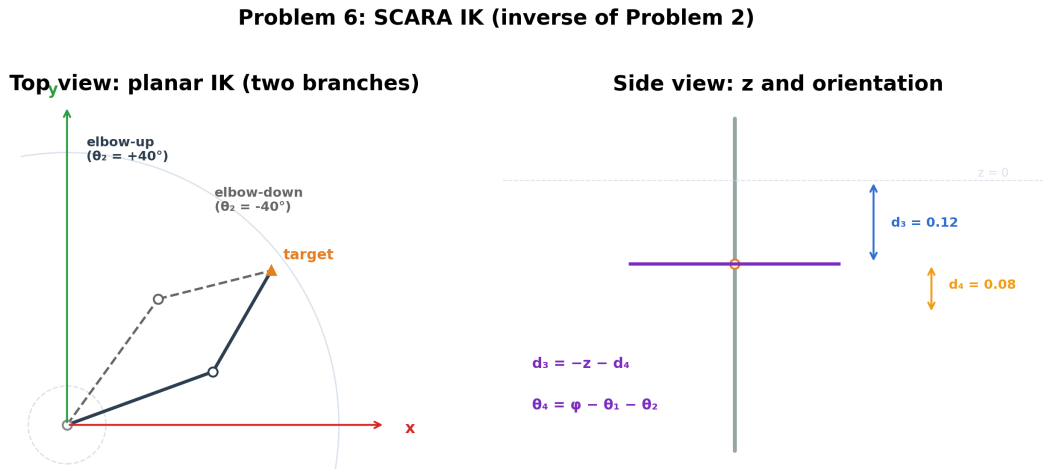


Figure 6: Problem 6 geometry

8.2 Solution

8.2.1 Reachability

$$r = \sqrt{0.5259^2 + 0.3966^2} = 0.6587 \text{ m}$$

Workspace: $|a_1 - a_2| = 0.10 \leq 0.6587 \leq 0.70 = a_1 + a_2$. Reachable.

8.2.2 Planar Sub-Problem

$$\cos \theta_2 = \frac{r^2 - a_1^2 - a_2^2}{2a_1a_2} = 0.7660 \quad \Rightarrow \quad \theta_2 = \pm 40^\circ$$

For each branch:

$$\theta_1 = \text{atan2}(y, x) - \text{atan2}(a_2 \sin \theta_2, a_1 + a_2 \cos \theta_2)$$

Elbow-up ($\theta_2 = 40^\circ$): $\theta_1 = 20.00^\circ$

Elbow-down ($\theta_2 = -40^\circ$): $\theta_1 = 54.04^\circ$

8.2.3 Vertical and Orientation

$$d_3 = -z - d_4 = 0.20 - 0.08 = 0.12 \text{ m}$$

$$\theta_4 = \phi - \theta_1 - \theta_2$$

Elbow-up: $\theta_4 = 30^\circ - 20^\circ - 40^\circ = -30^\circ$

Elbow-down: $\theta_4 = 30^\circ - 54.04^\circ + 40^\circ = 15.96^\circ$

8.2.4 FK Verification (Elbow-Up)

$$x = 0.40 \cos 20^\circ + 0.30 \cos 60^\circ = 0.5259 \checkmark$$

$$y = 0.40 \sin 20^\circ + 0.30 \sin 60^\circ = 0.3966 \checkmark$$

$$z = -(0.12 + 0.08) = -0.20 \checkmark, \quad \phi = 20 + 40 - 30 = 30^\circ \checkmark$$

This matches Problem 2 exactly.

8.2.5 Final Answers

	θ_1	θ_2	d_3	θ_4
Elbow-up	20.00°	40.00°	0.12 m	-30.00°
Elbow-down	54.04°	-40.00°	0.12 m	15.96°

8.3 Teaching Note

This problem creates a satisfying loop with Problem 2: the IK elbow-up solution recovers the exact configuration used in the FK problem. Point this out explicitly. Also compare with Problem 5: SCARA IK adds the z -equation (direct) and the orientation constraint (one more linear equation), but the core 2R sub-problem

is identical. Students should see that SCARA IK = planar IK + two direct equations.

9 Problem 7: 3R Elbow-Arm Inverse Kinematics

9.1 Statement

Use the same 3R elbow robot as Problem 4:

$$d_1 = 0.30 \text{ m}, \quad a_2 = 0.40 \text{ m}, \quad a_3 = 0.30 \text{ m}$$

Target:

$$\mathbf{p}_{target} = \begin{bmatrix} 0.4287 \\ 0.4287 \\ 0.3500 \end{bmatrix} \text{ m}$$

Find both arm solutions.

Problem 7: elbow-arm IK branches

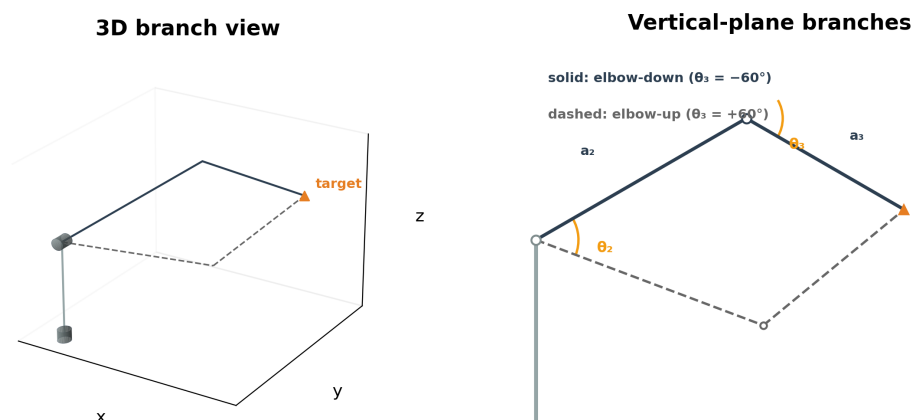


Figure 7: Problem 7 geometry

9.2 Solution

9.2.1 Step 1: Base Rotation

$$\theta_1 = \text{atan2}(y, x) = \text{atan2}(0.4287, 0.4287) = 45^\circ$$

9.2.2 Step 2: Reduce to a 2-Link Vertical-Plane Problem

$$r = \sqrt{x^2 + y^2} = 0.6062$$

$$s = z - d_1 = 0.3500 - 0.30 = 0.0500$$

Now the shoulder-elbow chain is a 2-link problem with lengths a_2 and a_3 .

9.2.3 Step 3: Solve for θ_3

$$\cos \theta_3 = \frac{r^2 + s^2 - a_2^2 - a_3^2}{2a_2a_3}$$

$$\cos \theta_3 = \frac{0.6062^2 + 0.05^2 - 0.40^2 - 0.30^2}{2(0.40)(0.30)} = 0.5$$

So:

$$\theta_3 = \pm 60^\circ$$

9.2.4 Step 4: Solve for θ_2

Use:

$$\theta_2 = \text{atan2}(s, r) - \text{atan2}(a_3 \sin \theta_3, a_2 + a_3 \cos \theta_3)$$

9.2.4.1 Case A: $\theta_3 = 60^\circ$ (elbow-up)

$$\theta_2 = \text{atan2}(0.05, 0.6062) - \text{atan2}(0.2598, 0.55) = 4.71^\circ - 25.28^\circ = -20.57^\circ$$

9.2.4.2 Case B: $\theta_3 = -60^\circ$ (elbow-down)

$$\theta_2 = \text{atan2}(0.05, 0.6062) - \text{atan2}(-0.2598, 0.55) = 4.71^\circ - (-25.28^\circ) = 30.00^\circ$$

9.2.5 FK Verification (Elbow-Down)

$$r = 0.40 \cos 30^\circ + 0.30 \cos(-30^\circ) = 0.6062$$

$$x = 0.6062 \cos 45^\circ = 0.4287 \checkmark, \quad y = 0.6062 \sin 45^\circ = 0.4287 \checkmark$$

$$z = 0.30 + 0.40 \sin 30^\circ + 0.30 \sin(-30^\circ) = 0.35 \checkmark$$

This matches the Problem 4 FK result.

9.2.6 Final Answers

- **Elbow-up:** $(\theta_1, \theta_2, \theta_3) = (45^\circ, -20.57^\circ, 60^\circ)$
- **Elbow-down:** $(\theta_1, \theta_2, \theta_3) = (45^\circ, 30^\circ, -60^\circ)$

9.3 Teaching Note

Students usually handle θ_1 correctly but lose the sign branch for θ_3 . Force them to write both branches explicitly before moving on to θ_2 . The elbow-down solution should recover the Problem 4 configuration — point this out to reinforce the FK/IK connection.

10 Problem 8: 6-DOF Spherical-Wrist Decoupling

10.1 Statement

Assume a 6-DOF robot with a spherical wrist and tool offset:

$$d_6 = 0.10 \text{ m}$$

Desired end-effector pose:

$$\mathbf{p}_{ee} = \begin{bmatrix} 0.4814 \\ 0.4263 \\ 0.4948 \end{bmatrix} \text{ m}$$

$$R_6^0 = \begin{bmatrix} 0.9498 & -0.3129 & 0.0014 \\ 0.3047 & 0.9256 & 0.2244 \\ -0.0715 & -0.2127 & 0.9745 \end{bmatrix}$$

Assume the first three joints have already been solved as:

$$\theta_1 = 40^\circ, \quad \theta_2 = 30^\circ, \quad \theta_3 = -50^\circ$$

with

$$R_3^0 = \begin{bmatrix} 0.7198 & -0.6428 & -0.2620 \\ 0.6040 & 0.7660 & -0.2198 \\ 0.3420 & 0 & 0.9397 \end{bmatrix}$$

Find:

1. the wrist center
2. the wrist rotation R_6^3
3. the wrist angles $\theta_4, \theta_5, \theta_6$

Problem 8: 6-DOF decoupling

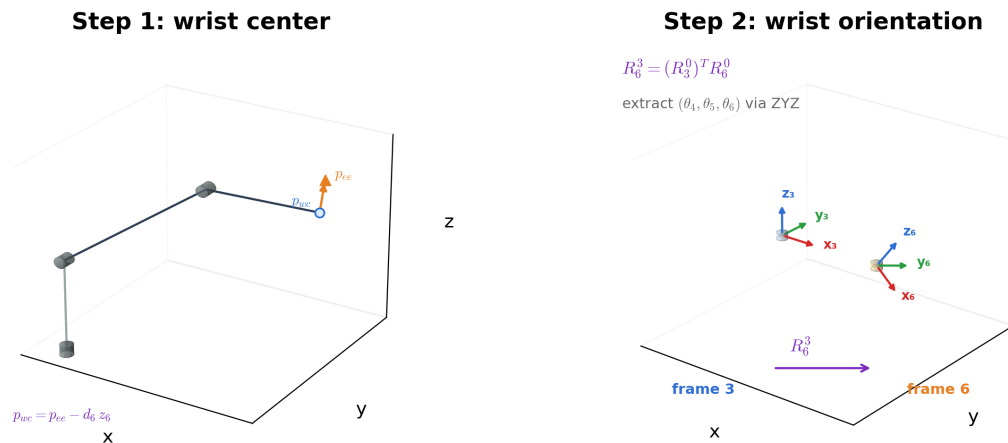


Figure 8: Problem 8 geometry

10.2 Solution

10.2.1 Step 1: Wrist Center

The third column of R_6^0 is the tool approach direction:

$$\mathbf{z}_6 = \begin{bmatrix} 0.0014 \\ 0.2244 \\ 0.9745 \end{bmatrix}$$

Then:

$$\mathbf{p}_{wc} = \mathbf{p}_{ee} - d_6 \mathbf{z}_6$$

$$\mathbf{p}_{wc} = \begin{bmatrix} 0.4814 \\ 0.4263 \\ 0.4948 \end{bmatrix} - 0.10 \begin{bmatrix} 0.0014 \\ 0.2244 \\ 0.9745 \end{bmatrix} = \begin{bmatrix} 0.4813 \\ 0.4039 \\ 0.3974 \end{bmatrix} \text{ m}$$

This is the point that joints 1 to 3 must place in space.

10.2.2 Step 2: Orientation Sub-Problem

Use decoupling:

$$R_6^3 = (R_3^0)^T R_6^0$$

Result:

$$R_6^3 = \begin{bmatrix} 0.8433 & 0.2611 & 0.4698 \\ -0.3771 & 0.9102 & 0.1710 \\ -0.3830 & -0.3214 & 0.8660 \end{bmatrix}$$

10.2.3 Step 3: Extract Wrist Angles

Using the same ZYZ extraction as Problem 3:

$$\theta_5 = \text{atan2}\left(\sqrt{0.4698^2 + 0.1710^2}, 0.8660\right) = 30^\circ$$

$$\theta_4 = \text{atan2}(0.1710, 0.4698) = 20^\circ$$

$$\theta_6 = \text{atan2}(-0.3214, 0.3830) = -40^\circ$$

10.2.4 Verification

Reconstruct: $R_z(20^\circ)R_y(30^\circ)R_z(-40^\circ)$ should match R_6^3 from Step 2. Then: $R_3^0 \cdot R_6^3$ should match R_6^0 .

10.2.5 Final Answer

$$\mathbf{p}_{wc} = \begin{bmatrix} 0.4813 \\ 0.4039 \\ 0.3974 \end{bmatrix} \text{ m}$$

$$(\theta_4, \theta_5, \theta_6) = (20^\circ, 30^\circ, -40^\circ)$$

10.3 Teaching Note

This is the capstone problem. If time runs short, it is still worth doing the wrist-center computation and setting up R_6^3 even if the full angle extraction is left for homework. Students who completed Problem 3's ZYZ extraction should recognize the identical procedure here.

11 Summary

11.1 Main Takeaways

1. Forward kinematics is a clean chain-building problem once the frames are assigned consistently.
2. Inverse kinematics becomes easier when the geometry is reduced before equations are solved.
3. Different robot types lead to different algebraic complexity:
 - SCARA: 2R planar IK + two direct equations
 - 3R elbow arms: base rotation + vertical-plane 2-link problem
 - spherical wrist: decoupled orientation recovery using ZYZ extraction
4. Every IK answer should be checked with a forward substitution.

11.2 FK/IK Pairings

Reinforce by explicitly connecting:

- Problem 2 $\rightarrow$ Problem 6 (SCARA)
- Problem 4 $\rightarrow$ Problem 7 (elbow arm)
- Problem 3 $\rightarrow$ Problem 8 (ZYZ wrist)

11.3 Suggested Use in Class

- Use the slide deck for pacing and reveal the solution slide only after students commit to a setup.
- Keep Problems 4 and 7 together because they reuse the same articulated-arm geometry.
- Problem 3's ZYZ extraction directly previews Problem 8, so reference it when you get there.
- If the class moves faster than expected, assign a numerical-IK comparison using the archived Week 5 material as an extension.